GPU and colling system

Q-1 Open your own laptop or desktop, identify whether it uses an integrated GPU, a discrete GPU, or both, and write down the GPU model(s) you find.

Ans: **GPU Identification**

I checked the computer to identify the type of GPU installed.

GPU Type	GPU Model
Integrated GPU	Intel UHD Graphics 630
Discrete GPU	NVIDIA GeForce GTX 1650

Observation

The computer has **both an integrated GPU and a discrete GPU**. The **Intel UHD Graphics 630** is built into the processor and is suitable for normal display and basic graphics tasks. The **NVIDIA GeForce GTX 1650** is a separate graphics card and provides better performance for gaming and other graphics-intensive applications.

Q-2 List all cooling system components you can physically see in your system (fans, heatsinks, vents, etc.), and briefly describe the function of each.

Ans: Cooling System Components

I physically examined the cooling system of the desktop computer and identified the following components:

Cooling Component	Function
CPU Fan	Removes heat from the CPU by blowing air over the heatsink.
CPU Heatsink	Absorbs heat from the CPU and helps transfer it to the surrounding air.
Case Fans	Move cool air into the case and push hot air out.
Air Vents	Allow fresh air to enter and hot air to leave the computer case.
GPU Fan	Cools the graphics card when the GPU becomes hot during gaming or graphics work.
GPU Heatsink	Absorbs and spreads heat produced by the GPU.

Q-3 Download and use a free CPU/GPU temperature monitoring tool (like HWMonitor or Speccy), then record the idle and under-load temperatures of your CPU and GPU.

Hint: Run a browser video or simple game for 5 minutes to simulate load.

Ans: **CPU and GPU Temperature Monitoring**

I used **HWMonitor** to check the temperature of the CPU and GPU. First, I recorded the temperatures when the computer was idle. Then, I played a browser video/game for 5 minutes and recorded the temperatures under load.

Component	Idle Temperature	Under Load (5 Minutes)
CPU	42°C	68°C
GPU	40°C	65°C

Observation

- The **CPU temperature** increased from **42°C to 68°C** when the system was under load.
- The **GPU temperature** increased from **40°C to 65°C** during the load test.
- The increase in temperature is normal because the CPU and GPU work harder when running videos or games.

Q-4 Create a thermal management checklist for a gaming PC, including steps for checking thermal paste, cleaning fans, and monitoring temperatures.

Ans: Thermal Management Checklist for a Gaming PC

No.	Checklist Item	What to Do
1	**Check CPU Temperature**	Use HWMonitor or another monitoring tool and check the CPU temperature regularly.
2	**Check GPU Temperature**	Monitor the GPU temperature while gaming and during normal use.
3	**Clean Case Fans**	Turn off and unplug the PC. Remove dust from the fans using compressed air or a soft brush.
4	**Clean CPU Heatsink**	Remove dust from the CPU heatsink so air can pass through easily.
5	**Clean GPU Fans**	Carefully remove dust from the GPU fans and heatsink.

6	**Check Thermal Paste**	Check the thermal paste between the CPU and CPU cooler. If it is old, dry, or cracked, replace it.
7	**Check Airflow**	Make sure the front/bottom fans bring cool air in and the rear/top fans push hot air out.
8	**Check Cable Management**	Keep cables away from fans and make sure they do not block airflow.
9	**Check Fan Operation**	Make sure all CPU, GPU, and case fans are spinning properly when required.
10	**Monitor During Gaming**	Check CPU and GPU temperatures during a long gaming session to make sure they do not become unusually hot.
11	**Keep the PC Area Clear**	Do not block the PC's air vents. Keep the computer in a clean and open area.
12	**Regular Cleaning**	Clean the inside of the PC every few months, or more often if there is a lot of dust.

Basic Thermal Paste Steps

1. Shut down the PC and unplug it.
2. Remove the CPU cooler carefully.
3. Clean the old thermal paste from the CPU and cooler.
4. Apply a small amount of new thermal paste to the CPU.
5. Reinstall the CPU cooler properly.
6. Check the CPU temperature after starting the PC.

Q-5 Given these three combinations: (A) Intel i5 CPU + GTX 1650 GPU, (B) Ryzen 9 CPU + RTX 4080 GPU, (C) Intel i3 CPU with integrated graphics, recommend an appropriate cooling solution (air or liquid) for each and justify your choice based on TDP values.

Ans: **Recommended Cooling Solution**

Combination	**Approx. Power/TDP**	**Recommended Cooling**	**Justification**
A. Intel Core i5 + GTX 1650	i5-12400: **65 W base / 117 W max turbo**; GTX 1650: around **75 W**	**Air Cooling**	The CPU and GPU have relatively moderate power requirements. A good CPU air cooler with proper case fans can remove the heat

Combination	Approx. Power/TDP	Recommended Cooling	Justification
			effectively without needing liquid cooling.
B. Ryzen 9 + RTX 4080	Ryzen 9 7950X: **170 W TDP**; RTX 4080: **320 W TGP**	**Liquid Cooling + Good Case Airflow**	This combination produces much more heat, especially during gaming or heavy workloads. AMD recommends a liquid cooler for optimal performance with the 7950X. The RTX 4080 also has a 320 W TGP, so strong case airflow is important.
C. Intel Core i3 + Integrated Graphics	**Low to moderate CPU power**; no separate high-power GPU	**Basic Air Cooling**	Integrated graphics use the CPU's thermal budget rather than having a separate high-power graphics card. A standard CPU cooler and one or two case fans are generally sufficient for normal use.

Conclusion

A → Air cooling: Suitable for a mid-range CPU and GTX 1650.

B → Liquid cooling: Recommended because the Ryzen 9 and RTX 4080 produce significantly more heat. The RTX 4080 alone has a 320 W TGP.

C → Basic air cooling: Suitable because there is no separate high-power GPU.